



MINING BEST PRACTICES

Road Analysis Control (RAC)

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

This Best Practice provides an overview of the Road Analysis Control (RAC) tool and best practices for a successful implementation of RAC as a haul road management and productivity tool.

RAC is an on-board information technology product designed to measure and benchmark haul road quality. Integrated with the Vital Information Management System (VIMS), RAC measures component loading and impact shock, communicates that data to the operator and to the mine office in real time. Trucks equipped with GPS can export a RAC event location so the events can be plotted on the mine map and proactively creating a work plan for the mine's road maintenance team. RAC provides data and reports through Technician Toolbox and enables mine managers to quantify, monitor and manage haul roads.

2.0 Opportunity Recognition

The best opportunities for RAC implementation can be found in mines with:

- Poor tire life
- Frame, lower powertrain, and strut quality issues – especially frame downtime
- Operator complaints of poor ride and haul road quality
- Inconsistent cycle times day-to-day
- Underfoot prone to pits
- Haul road maintenance repair is determined by operator feedback

3.0 Benefits

Road Analysis Control (RAC), is a haul road management tool providing feedback to help manage operator technique, haul road and loading conditions.

This feedback enables the site to, quantify the severity of the condition and allows mine management to make more informed decisions to reduce overall haulage costs.

The long-term benefits of using RAC can include tire life improvements as well as the same benefits mine's receive from good haul road maintenance – haul trucks move faster and have less issues when they travel along smooth, well maintained haul roads. Faster trucks increase mine production, having less maintenance issues on trucks reduces haulage costs so incorporating RAC into mine operations improves mine cost-per-ton.

When used properly RAC can be a significant productivity tool to:

- Improve haulage cycle times and raise truck productivity.
- Contribute to longer tire and rim life.
- Identify haul road problem areas to avoid and correct.
- Lower operating costs. Help achieve intended component life cycle.
- Identify haulers with poorly charged suspension
- Improve repair before failure processes.
- Reduce stress on drive components, frame and suspension.
- Reduce fuel consumption.

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- Provide component life and repair indicator information.
- Give haul road support equipment team specific locations for road repairs.
- Eliminate the human error where operators may not report road quality issues.

4.0 Resources Required

- Haul road maintenance supervisor and team
- Truck maintenance supervisor
- Dealer support – collect data, work through implementation, and follow up on progress.

5.0 Setup

Installation:

RAC is a Software Control Application (SCA) within Caterpillar Electronic Technician (ET) that can be activated by authorized dealer personnel. Machines must be equipped with GPS to pin-point locations on the haul roads that triggered RAC events. The mapping of this events can be reviewed within Health Equipment Insights (HEI).

RAC has 3x severity levels:

- Level 3 – 170% of target level 1 (least sensitive)
- Level 2 – 135% of target level 2
- Level 1 – baseline (most sensitive)

Recommendation is to start a mine site at Level 3 (least sensitive). If they're not getting events, they should move down to level 2 (more sensitive) and then onto level 1 (most sensitive). The mine does not start at Level 3, the number of events is likely to be difficult to handle.

At this point, it's also important to make sure the strut pressures are maintained properly and reference SEHS9411 document; otherwise, you'd get poor results from RAC.

It's recommended that all trucks activate RAC to monitor to collect the most data.

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6.0 Haul Road Maintenance

Events can be collected during shifts or for multiple days, a recommended length of time to collect is one week, in order to get enough data to overlapping in the critical areas.

After collecting the data for one reporting cycle, the team should overlay the events on a map of the entire mine. Because the RAC data is tagged with GPS location, the mine can use the latest mine maps to overlay the events to their exact location. Ideally, there's a dispatch / production management system which could be MineStar Edge or another system. When the events come off the truck, they go to this system. From there they can do an event plot which takes all the events and plot them on a mine map.

After overlying the events, because the size of a mine can be quite significant, it's recommended that the events be grouped different zones (see figure 1) to help focus in on those areas. This example shows the bias type events in orange and rack type events in green.



Figure 1 – Event entire mine overlay

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Once separated into zones, the team can create a plan for road maintenance. Using the example shown, here is an example of focused in on zone 2 (see Figure 2). The maintenance team can clearly see that there are events on and off the haul roads that need repaired first.



Figure 2 - Zone 2 zoom-in

Events greater than (+/-) 3700 (unitless) should also be addressed first by the road maintenance team, as those could lead to frame and suspension durability issues. Events of that magnitude are generally mean that tires came off the ground and frame, suspension, drivetrain inspections must be performed at the next PM.

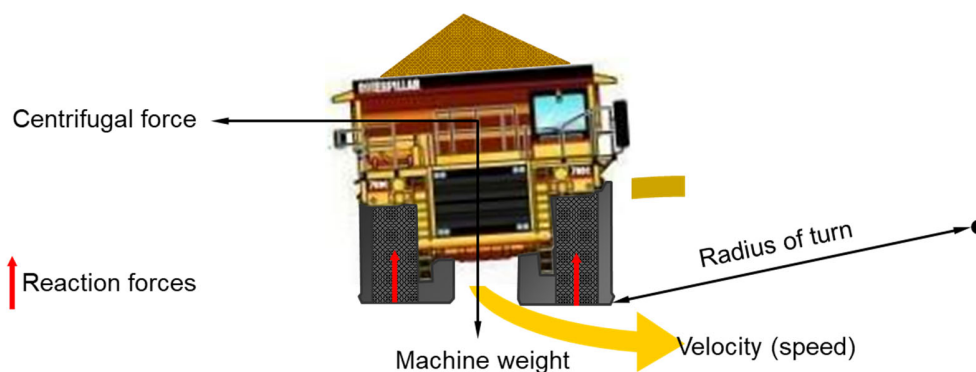
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7.0 Tire Management

There are several common areas where RAC can help support a good tire management program. The operators need to be trained on each of the unique RAC events and the conditions that create the event.

7.1 Common Bias Events - Cornering

The most common bias event is observed where the operator is steering a truck through a corner. If the cornering speed is too high the load of the truck will transfer to the outer tires and the tires will experience a lateral loading at the same time. The combination of these loads can be extremely stressful on the tire construction and often lead to shoulder separation issues on the tire.



7.2 Common Bias Events – Bias Tire Loading

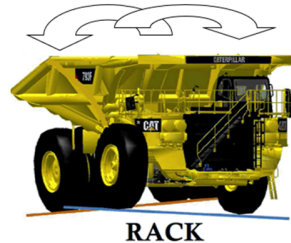
Another common bias event often seen is where the truck is approaching the dump. In this example the operator is applying the brakes (transferring load to the front tires) and simultaneously turning the truck. These examples have observed tire loads to reach over 180% of equivalent static loads while cornering. These events often result in a simultaneous rack event.

Coaching the operators to remove most of the speeds in a straight line and then cornering at a constant speed will under normal circumstances reduce the loads on the front tires by around 30 to 40%. Also, the team can instructor the motor grader to increase the cross fall angle or supper elevation on this turn to handle additional speed.

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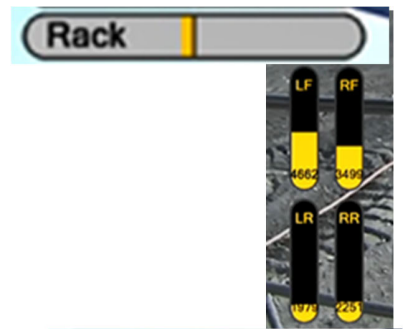
7.3 Common Rack Events – Bias Tire Loading

Rack events are very common while under the loading equipment and are effectively an indication of how significantly the loads are being applied to ‘twist’ the frame.

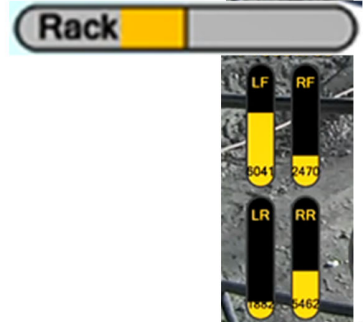


The following sequence of events is commonly seen on many operations during the loading phase:

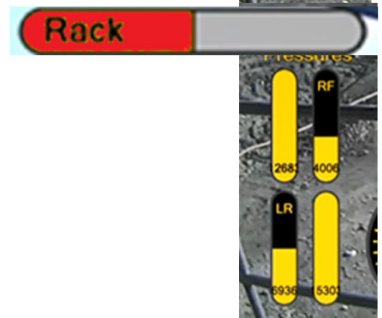
Prior to reversing in to the loading tool, The suspension pressures are very consistent left to right suggesting that they are well charged. Rack values are close to zero at this time.



As the truck approaches the loading position it is possible to observe when one of the duals is on the rilled material of the bench. This could be a single rock that could be a significant point loading on the tire or alternatively or the toe which will create frame stresses. In this position while empty it will not create a Rack event and is not creating significant harm to the tires/frame



As subsequent passes from the loading tool are added to the truck the weight will increase and subsequently the stresses across the machine can increase to the level that a Rack event will be induced.



It is worth noting that if there is a single rock underneath one of the duals that is inducing this situation it is reasonable to consider that 2/3 of the Gross vehicle mass (rear of truck with load) is largely being point loaded on the tire.

Rack events under the loading tool can easily be avoided by moving forward by a few meters between the first and second loading tool pass with negligible production implications but very significant benefits. Another best practice is when the truck is positioned under the shovel, instruct the operator to place the truck in Neutral, release the foot brake, and let the truck settle into a flat location before applying the brake again.

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8.0 Follow-up and Sustaining

Once the RAC and haul road maintenance plan has been implemented, the dealer and customer should schedule a follow-up to review the progress and improvements. The teams should keep track of the events logged over time and see a reduction in events over time if the road maintenance team is following the reports. It's important that the data and recommendations made to the haul road team be confirmed as completed and the process be implemented permanently, as to not go back to haul road conditions as before.

9.0 Supporting Attachments / References

ECM requirements:

- VIMS™ ABL or later must be present on the machine.
- Machines equipped with VIMS™ ABL required an additional ECM installed.
 - Reference document (SEHS0394)
- Machines equipped with VIMS™ 3G and PLE do not require an additional ECM.
- Installation of RAC on a Large Mining Truck with VIMS™
 - Reference document (REHS0875)

Enabling RAC:

- Authorization to enable the Software Control Application (SCA) within Caterpillar Electronic Technician (ET). Within CAT ET, authorized personnel can activate the SCA for RAC and will be invoiced for each machine.
 - Reference document (REHS7342)

10.0 Related Best Practices

1106-1.1-1035	Cost Reduction Through Haul Road Management Process
1107-1.2-1094	Road Defect Management Training Exercise
0409-2.02-1165	Haul Road Optimization

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